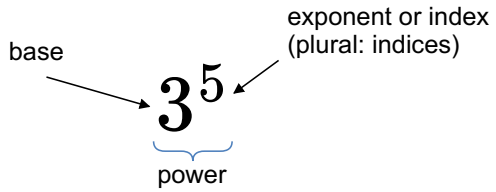
1.01 Indices

Chapter 1 · Original slide 4

1: Index Laws

4



$$a^m \times a^n = a^{m+n}$$
$$a^m \div a^n = a^{m-n}$$
$$(a^m)^n = a^{mn}$$
$$(ab)^n = a^n b^n$$

Some also refer to the 5 as the ‘power’. But this would lead to unfortunate ambiguous phrases like “when we multiply two powers together, we add the powers” instead of “when we multiply two powers together, we add the indices”.

Simplify $(a^3)^2 \times 2a^2$

Simplify $(4x^3y)^3$

Simplify $2x^2(3+5x) - x(4-x^2)$

Simplify $\frac{x^3-2x}{3x^2}$

Chapter 1 · Original slide 5

Test Your Understanding

5

1 Simplify $\left(\frac{2a^5}{a^2}\right)^2 \times 3a$

2 Simplify $\frac{2x+x^5}{4x^3}$

3 Expand and simplify
$2x(3-x^2) - 4x^3(3-x)$

4 Simplify $2^x \times 3^x$

Note: This is using $(ab)^n = a^n b^n$ law backwards.

1.01 Indices

Chapter 1 · Original slide 6

Exercise 1A – Pearson Pure Y1 page 4

6

2 Expand and simplify if possible:

o $x(3x^2 - 2x + 5)$
p $7y^2(2 - 5y + 3y^2)$
q $-2y^2(5 - 7y + 3y^2)$
r $7(x - 2) + 3(x + 4) - 6(x - 2)$
s $5x - 3(4 - 2x) + 6$
t $3x^2 - x(3 - 4x) + 7$
u $4x(x + 3) - 2x(3x - 7)$
v $3x^2(2x + 1) - 5x^2(3x - 4)$

3 Simplify these fractions:

d $\frac{8x^3+5x}{2x}$ **e** $\frac{7x^7+5x^2}{5x}$ **f** $\frac{9x^5-5x^3}{3x}$

Chapter 1 · Original slide 7

Exercise 1A – Pearson Pure Y1 page 4

7

Extension (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1 *[MAT 2006 1A]*

Which of the following numbers is largest?

- $\left(\left(2^3\right)^2\right)^3$
- $\left(2^3\right)^{\left(2^3\right)}$
- $2^{\left(\left(3^2\right)^3\right)}$
- $2^{\left(3^{\left(2^3\right)}\right)}$

2 *[MAT 2012 1B]*

Let $N = 2^k \times 4^m \times 8^n$ where k, m, n are positive whole numbers.

Then N will definitely be a square number whenever:

- k is even;
- $k + n$ is odd;
- k is odd but $m + n$ is even;
- $k + n$ is even.

1.01 Indices

Chapter 1 · Original slide 17

4: Negative and Fractional Indices

17

$a^0 = 1$
$a^{\frac{1}{m}} = \sqrt[m]{a}$
$a^{\frac{n}{m}} = \left(\sqrt[m]{a}\right)^n$
$a^{-m} = \frac{1}{a^m}$

Note: $\sqrt{9}$ only means the positive square root of 9, i.e. 3 not -3.
Otherwise, what would be the point of the $\pm$ in the quadratic formula before the $\sqrt{b^2 - 4ac}$?

Prove that $x^{\frac{1}{2}} = \sqrt{x}$

Evaluate $27^{-\frac{1}{3}}$

Evaluate $32^{\frac{2}{5}}$

Simplify $\left(\frac{1}{9}x^6y\right)^{\frac{1}{2}}$

Evaluate $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$

If $b = \frac{1}{9}a^2$, determine $3b^{-2}$ in the form kb^n where k, n are constants.

Chapter 1 · Original slide 18

Writing a surd using indices

18

If $9\sqrt{3} = 3^k$, determine the value of k.

The key here is to write everything as powers with a consistent base, in this case, 3.

Note: In algebra we like to avoid mixed numbers. So don't write $3^{2\frac{1}{2}}$

1.01 Indices

Chapter 1 · Original slide 19

Further Examples

19

[Edexcel IGCSE May14-4H]
Given that

$$\left(2^{\frac{1}{2}}\right)^{n} = \frac{2^{x}}{8^{y}}$$

Express n in terms of x and y.

[Edexcel IAL C12 Jan 2019 Q2c] Given that $y = 2^{x}$, express the following in terms of y.

$$\frac{1}{4^{2x-3}}$$

Write your expression in its simplest form.

Write 8 as a power of 2 (putting a bracket around it), for consistency of base with the other powers.

Chapter 1 · Original slide 20

Test Your Understanding

20

Edexcel Paper 2 – May 2019

Given $2^{x} \times 4^{y} = \frac{1}{2\sqrt{2}}$ express y as a function of x. (3)

1.01 Indices

Chapter 1 · Original slide 21

Exercise 1D

21

4 a Find the value of $81^{\frac{1}{4}}$. **b** Simplify $x(2x^{-\frac{1}{3}})^4$.

5 Given that $y = \frac{1}{8}x^3$ express each of the following in the form kx^n, where k and n are constants. **a** $y^{\frac{1}{3}}$ **b** $\frac{1}{2}y^{-2}$

Challenge

[MAT 2007 1A]

Let r and s be integers. Then

$$\frac{6^{r+s} \times 12^{r-s}}{8^r \times 9^{r+2s}}$$

is an integer if

- $r + s \leq 0$
- $s \leq 0$
- $r \leq 0$
- $r \geq s$